

IgG4-related disease of the central and peripheral nervous systems



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IgG4-related disease can involve nearly any organ system, including the central and peripheral nervous systems. The pathology findings are consistent from organ to organ, but careful clinicopathological correlation is necessary to establish the diagnosis. Many non-neurological and neurological inflammatory conditions, previously regarded as idiopathic in nature, are now recognised to fall within the spectrum of IgG4-related disease. The condition is highly treatable, but probably remains substantially under-recognised. In this Review, we offer an important and timely update on the current and emerging aspects of this neurological disease. Following a short overview of IgG4-related disease, we describe the current understanding of neurological findings, pathophysiology, approaches to diagnosis, and treatment of IgG4-related disease affecting the central and peripheral nervous systems.

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Introduction

IgG4-related disease (IgG4-RD) was first recognised as a discrete, immune-mediated disease in 2003.¹ IgG4-RD can affect multiple organ systems, including both the central and peripheral nervous systems. The principal neurological manifestations result from orbital disease, pachymeningitis, and pituitary gland and stalk involvement, but peripheral nerve disease and parenchymal brain infiltrates have also been reported. The clinical manifestations of IgG4-related neurological disease frequently mimic malignancy, infection, and other inflammatory diseases, thereby confounding diagnostic efforts. IgG4 antibodies themselves are not believed to have a primary role in disease aetiology. The disease must therefore be distinguished through the correlation of clinical features with pathological, serological, and radiological findings from other immune-mediated diseases in which IgG4 auto-antibodies contribute directly to pathophysiology. IgG4-RD is highly treatable with either glucocorticoids or targeted biological drugs, particularly those that deplete B cells, and consensus therapeutic approaches have been formulated by a collaboration of multi-specialty experts from Europe, North America, and Asia.

Epidemiology

Defining the epidemiology of IgG4-RD has been hampered by insufficient awareness of this relatively new disease.¹ Definitive diagnosis of IgG4-RD generally requires a tissue sample, insightful interpretation of the pathology, and rigorous clinicopathological and clinicoradiological correlation, all of which add complexity to epidemiological studies. IgG4-RD was first recognised in Japan as a disease affecting the pancreas.² The prevalence of IgG4-related autoimmune pancreatitis increased from 2.2 cases per 100 000 population in 2007 to 4.6 cases per 100 000 population in 2011, an increase almost certainly due to greater recognition.^{3,4} These data certainly underestimate the true worldwide prevalence of IgG4-RD because many doctors are unaware of the disease, and the mechanisms for reporting disease for the purposes of epidemiological studies remain inadequate for many

disease manifestations. Moreover, the pancreas is only one of more than a dozen organs affected by IgG4-RD.^{5,6}

The typical patient with systemic IgG4-RD is a middle-aged to elderly man. The mean age at diagnosis is 55–59 years, and the male-to-female ratio ranges from 1.0:0.64 to 1.0:0.77.^{7,8} Disease severity does not seem to vary according to sex.⁸ The epidemiology of IgG4-RD of the nervous system generally reflects that of the condition overall. IgG4-related pachymeningitis is a disease of middle-aged or elderly individuals (mean age 53 years), and about 65% of the cases reported to date are in men.⁹ IgG4-related hypophysitis generally presents at the seventh decade of life (mean age 62 years) and is associated with a 3.6:1 male predominance,^{10,11} by contrast with the more common lymphocytic hypophysitis, which has a female predominance and tends to present in the fourth decade.¹² IgG4-related orbital disease is more balanced in terms of sex distribution,^{13–17} and the mean age at diagnosis is 50 years in Europe^{14–17} and 61 years in Japan.¹³

Pathophysiology

Much has been learned about the immunology of IgG4-RD in a short period. The condition seems to be an antigen-driven process, albeit the antigen (or antigens) initiating the process remain unknown. Crucial interactions occur between cells of the B-cell lineage (both B lymphocytes and plasmablasts) and two CD4+ T lymphocytes, the T-follicular helper (CD4+ Tfh) cells and CD4+ cytotoxic T lymphocytes (CD4+ CTL). This basic immunological understanding illuminates a number of potential targets for therapy. Patients with IgG4-RD have increased plasmablasts in the peripheral blood, even in the setting of a normal serum IgG4 concentration.¹⁸ The most abundant cells at sites of tissue involvement, however, are CD4+ T cells.¹⁹

An overview of the pathophysiology is shown in figure 1. Two major interactions between cells of the B-lymphocyte lineage and CD4+ T lymphocytes drive the pathophysiology of IgG4-RD.²⁰ The first interaction includes antigen presentation by both B cells (CD20+) and plasmablasts (CD20–CD19+CD38+CD27+) to the CD4+ CTL. An oligoclonally expanded population of CD4+ CTLs in both

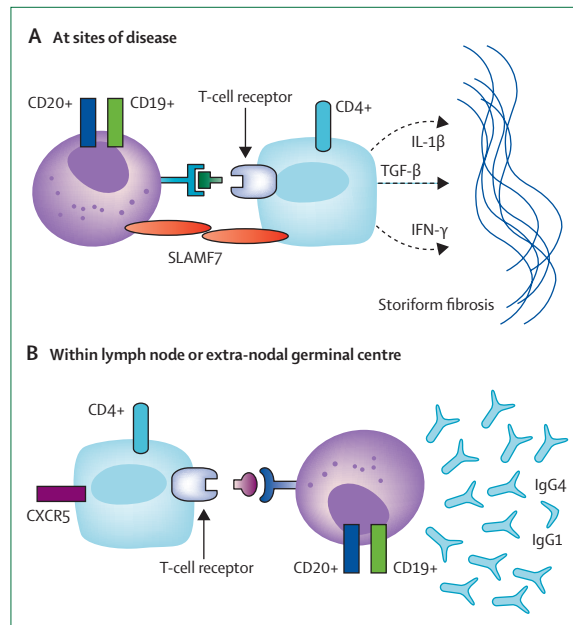


Figure 1: Overview of the pathophysiology of IgG4-related disease

Two major interactions between cells of the B-lymphocyte lineage and CD4+ T-lymphocytes appear to drive the pathophysiology of IgG4-related disease. (A) An antigen the identity of which remains unknown and might vary between patients is presented by B cells and plasmablasts to a CD4+ cytotoxic T lymphocyte, which is presumably sustained by continuous antigen presentation and elaborates not only fibrogenic cytokines (eg, IL-1 β , IFN- γ , and TGF- β) but also cytotoxic mediators such as granzyme B and perforin. SLAMF7, a molecule present on both cells, could be the means by which the two cells adhere to each other. (B) T-follicular helper 2 cells, also CD4+, residing within lymph node germinal centers and in secondary and tertiary lymphoid organs at disease sites, drive B cells to switch to IgG4 production (ie, class switching). The function of IgG4 in this setting remains uncertain but might constitute an ineffectual counter-regulatory mechanism. Class-switched plasmablasts return to the bone marrow, where they might survive as long-lived plasma cells, continuing to secrete IgG4. CXCR5 is a molecule that identifies the cell as a T-follicular helper cell. Both IgG4 and IgG1 are produced by activated cells of the B-lymphocyte lineage, but IgG4 production is increased out of proportion to IgG1. IL-1 β =interleukin-1 β . TGF- β =transforming growth factor β . IFN- γ =interferon- γ .

the peripheral blood and fibrotic lesions of patients with IgG4-RD suggests that these cells are central to the disease.²¹ These CD4+ CTLs secrete interleukin-1 β , interferon- γ , and transforming growth factor β , all of which might contribute to the storiform fibrosis, characteristic of IgG4-RD. The CD4+ CTLs also actively secrete products that are associated more commonly with CD8+ T cells, such as granzyme B and perforin.

Current thinking about the pathophysiology of IgG4-RD holds that the CD4+ T cells orchestrating the disease are sustained by continuous antigen-presentation by B cells and plasmablasts. The rapidity with which the serum IgG4 concentration decreases after B-cell depletion by anti-CD20 treatment suggests that plasmablasts and short-lived plasma cells make most of the serum IgG4.²² However, if the serum IgG4 concentration is extremely high before treatment, it typically does not normalise for many months or longer (and might never normalise),

suggesting that a sizeable proportion of serum IgG4 is derived from long-lived plasma cells (once circulating plasmablasts) that have homed back to the bone marrow and established long-term niches.²³

Circulating plasmablasts in patients with IgG4-RD undergo intense somatic hypermutation,²³ a hallmark of interaction with T cells within the germinal centres of lymph nodes. This process reflects the second characteristic interaction between cells of the B-lymphocyte lineage and CD4+ T cells, which occurs both within lymph node germinal centres and in secondary and tertiary lymphoid organs at disease sites to drive the class switch, whereby B cells secrete IgG4 in preference to other IgG subclasses.²⁴ For reasons that remain unknown, the CD4+ Tfh cells drive the class switch (by secreting interleukin-4) disproportionately toward the production of IgG4. The particular subset of CD4+ Tfh cells responsible for this class switch appears to be T-follicular helper 2 cells.¹⁹ It is now recognised that T-follicular helper cells (CD4+ Tfh cells) are subdivided into at least three types termed Tfh1, Tfh2, and Tfh3 cells. The Tfh2 subset seems to be crucial in IgG4-RD. The site of this interaction seems to be at the periphery of germinal centres.

Ironically, IgG4 might have only a counter-regulatory mechanism in disease pathophysiology.¹⁹ This hypothesis is appealing because IgG4 is known to bind with low affinity to complement and Fc receptors on effector immune cells (eg, macrophages and natural killer cells) and is thus a weak inducer of antibody-dependent effects such as cellular cytotoxicity, cellular phagocytosis, and complement activation.²⁵ The purpose of IgG4 in IgG4-RD might be to serve as an antigen sink at sites of inflammation, binding loose antigen and thereby tempering rather than exacerbating the existent inflammatory locus.

Distinguishing IgG4-RD of the nervous system from neurological diseases mediated by IgG4 antibodies

IgG4-RD must be distinguished from a growing number of neurological diseases in which IgG4 autoantibodies have a direct role in pathophysiology.²⁶ In these disorders, the IgG4 subclass is the predominant antibody subclass formed against the target molecule. Certain patients with chronic inflammatory demyelinating polyneuropathy, for example, have pathogenic IgG4 antibodies against contactin 1, the contactin 1-associated protein-1,²⁷ or neurofascin-155,²⁸ all of which are protein constituents within the nodes of Ranvier of peripheral nerves. In myasthenia gravis, the IgG4 subclass has been repeatedly shown to have the most pathogenicity against muscle-specific kinase.^{29,30} Finally, leucine-rich glioma inactivated 1 (LGI1) antibodies, which are linked to the newly described phenotype of limbic encephalitis (characterised by faciobrachial dystonic seizures), are predominantly of the IgG4 subclass, with occasional IgG1 anti-LGI1 molecules seen in some patients.³¹ By contrast with this group of

neurological disorders, IgG4 antibodies are not known to have a primary role in the disease aetiology of IgG4-RD.

Overview of systemic IgG4-RD

General features and common clinical presentations

The non-neurological organ system manifestations of IgG4-RD are summarised in the table. The presentation of IgG4-RD is typically subacute, with symptoms and signs present for months or even years before the diagnosis. Weight loss of 5–15 kg can occur over months, but rampant symptoms of inflammation, such as fevers, are unusual. Fatigue commonly accompanies IgG4-RD, particularly when the disease affects multiple organ systems. Many patients have arthralgias and other musculoskeletal symptoms, particularly enthesopathy (inflammation at the site of a tendinous insertion into a bone), but frank arthritis is atypical and suggests other diseases.

Two clinical presentations are common. First, IgG4-RD might present as a mass lesion, mimicking malignancy, in the submandibular gland, pancreas, or kidney, or in the pituitary gland or meninges.⁶ Second, many patients develop symptoms of allergy or atopy. Allergic rhinitis, nasal polyps, chronic sinusitis, nasal obstruction, and rhinorrhoea are common in IgG4-RD. Mild to moderate peripheral eosinophilia and serum IgE concentration that sometimes exceeds ten times the upper limit of normal are also common.³²

Organ involvement beyond the nervous system

Lacrimal gland enlargement is the most common feature of ophthalmic disease in IgG4-RD. The triad of dacryoadenitis and enlargement of both the parotid and submandibular glands, known for more than 100 years as Mikulicz' disease,³³ has a high positive predictive value for the diagnosis of IgG4-RD. Isolated submandibular gland enlargement is also a common and highly distinctive disease feature.

The lymphadenopathy associated with IgG4-RD is typically either a generalised lymphadenopathy or localised disease adjacent to an affected organ.³⁴ The lymph nodes involved are generally 1–2 cm in diameter and non-tender. Lymph node biopsies are not recommended for establishing the diagnosis of IgG4-RD because the findings have poor specificity.

The most common clinical presentation of IgG4-related autoimmune pancreatitis is obstructive jaundice, caused by swelling of the organ and compression of the pancreatic duct. Endocrine or exocrine failure of the pancreas can lead to diabetes mellitus, malabsorption, and weight loss.³⁵ Autoimmune pancreatitis is often associated with IgG4-related sclerosing cholangitis, affecting both the intra-hepatic and extra-hepatic bile ducts.

About two-thirds of cases of idiopathic retroperitoneal fibrosis are caused by IgG4-RD.³⁶ IgG4-RD targets three general anatomical locations in this area: periaortic or arterial regions; periureteral areas; and a plaque-like mass that broadly involves the retroperitoneum. IgG4-related

Clinical presentations	
Ears, nose, and sinuses	Allergic phenomena (nasal polyps, asthma, allergic rhinitis, peripheral eosinophilia); anosmia, chronic sinusitis; occasional bone-destructive lesions
Salivary glands	Submandibular or parotid gland enlargement, or both (isolated bilateral submandibular gland involvement is the most common)
Lymph nodes	Either generalised lymphadenopathy (1–2 cm, non-tender) or localised disease adjacent to a specific affected organ
Thyroid gland	Riedel's thyroiditis: thyroid enlargement and extension through tissue plane, causing neck pain, anterior neck swelling, dyspnoea, dysphagia, and dysphonia
Lungs	Asymptomatic finding on lung imaging; cough, haemoptysis, dyspnoea, pleural effusion, or chest discomfort associated with parenchymal lung involvement, pleural disease, or both; four main clinical syndromes are inflammatory pseudotumour, central airway disease, localised or diffuse interstitial pneumonia, and pleuritis; pleural lesions have severe, nodular thickening of the visceral or parietal pleura with diffuse sclerosing inflammation, sometimes associated with pleural effusion
Aorta	Asymptomatic finding on radiological studies; unanticipated discovery at elective aortic surgery, and aortic dissection; clinicopathological syndromes described include lymphoplasmacytic aortitis of thoracic or abdominal aorta; aortic dissection; peri-aortitis and periarteritis; inflammatory abdominal aneurysm
Retroperitoneum	Poorly localised pain in the back, flanks, lower abdomen, or thighs; lower extremity oedema; hydronephrosis from ureteral involvement; asymptomatic finding on radiological studies
Kidney	Proteinuria, haematuria, decreased kidney function, and radiological abnormalities; typical renal syndrome is tubulointerstitial nephritis; nephrotic range proteinuria occurs in a small proportion of patients and, when present, is associated with membranous glomerulopathy; either subacute or progressive chronic renal failure can occur; asymptomatic tumoral lesions, typically multiple and bilateral, are sometimes detected on radiological studies; many patients with IgG4-RD and renal disease are hypocomplementaemic, sometimes profoundly so
Pancreas	Mild abdominal pain, weight loss, and acute, obstructive jaundice mimicking adenocarcinoma of the pancreas (including a pancreatic mass); 20–50% of patients present with acute glucose intolerance; a small proportion of patients present with either acute pancreatitis (severe upper abdominal pain, lipase elevation) or chronic pancreatitis with features of exocrine insufficiency and pancreatic calcification, or pseudocysts
Biliary tree	Obstructive jaundice associated with autoimmune pancreatitis in most cases, weight loss, steatorrhoea, abdominal pain, and new-onset diabetes mellitus; mimics primary sclerosing cholangitis
Liver	Mass lesion of the liver; obstructive jaundice
Other organs	Gallbladder (lymphoplasmacytic sclerosing cholecystitis); breast (pseudotumour); prostate; pericarditis (constrictive pericarditis); stomach; sclerosing mesenteritis; sclerosing mediastinitis; skin (erythematous or flesh-coloured papules)
IgG4-RD=IgG4-related disease.	
Table: Summary of major manifestations of IgG4-RD beyond the nervous system and orbits	

aortitis can also lead to aneurysms or dissections in the thoracic aorta. By contrast with giant cell and Takayasu's arteritis, however, IgG4-RD tends to spare the primary aortic branches, particularly the subclavian arteries.

Pulmonary nodules, ground-glass opacities, pleural thickening, and interstitial lung disease are all common radiological features of IgG4-RD.³⁷ The interstitial lung disease mimics non-specific interstitial pneumonitis and other forms of interstitial fibrosis. Thickening of the bronchovascular bundle, best appreciated on CT examination, underscores the tendency of IgG4-RD to track along bronchi and blood vessels.

Tubulointerstitial nephritis is the most characteristic form of IgG4-related kidney disease, but cases of membranous glomerulonephropathy have also been described.³⁸ Advanced renal dysfunction can result despite benign-appearing urinalyses and urine sediments.

Neurological IgG4-RD

IgG4-RD affects the nervous system by one of two general mechanisms. First, IgG4-related inflammation can infiltrate directly into the substance of the central or peripheral nervous system. Second, neurological structures can be compressed by the mass effect of nearby diseased non-neurological organs. In order of decreasing frequency, IgG4-RD can manifest in the substance of the dura matter (ie, pachymeninges), the pituitary gland and stalk (hypophysitis), the peripheral nerves, and the brain parenchyma. IgG4-related orbital disease also often presents with neurological symptoms. Occasionally, IgG4-RD might occur in the substance of the pia matter (ie, leptomeninges) without pachymeningeal involvement.^{39,40} Locations that have not yet been found to be affected by IgG4-RD include the parenchyma of the spinal cord, the dorsal root ganglia, the brachial and lumbo-sacral plexuses, and skeletal muscles outside the orbit and head.

Neurological sequelae of IgG4-related orbital disease (clinical and imaging features)

IgG4-related orbital disease represents about a fifth of all inflammatory or malignant disorders that affect the orbit.^{13,41} In the past, such orbital lesions were broadly labelled as idiopathic orbital inflammation, often without a biopsy or appropriate staining when a biopsy was done (such as staining for IgG4-RD). The differential diagnosis of IgG4-related orbital disease is broad and includes sarcoidosis, granulomatosis with polyangiitis, Graves' disease, lymphoma, and infections of the orbital tissue. Structures within the orbit that show neurological symptoms if affected by IgG4-RD include: the extra-ocular muscles and levator palpebrae; structures emerging from the optic canal such as the optic nerve (cranial nerve [CN] II) and ophthalmic artery; structures emerging from the superior orbital fissure as the superior and inferior branches of the oculomotor nerve (CN III) or the trochlear (CN IV), frontal (CN V1), lacrimal (CN V1), or nasociliary (CN V1) nerves; and structures emerging from the inferior orbital fissure as the infra-orbital (CN V2) and zygomatic (CN V2) nerves. IgG4-related orbital disease can potentially involve any of these structures with or without involving the orbital soft tissue itself (figure 2) or the lacrimal gland and its duct.

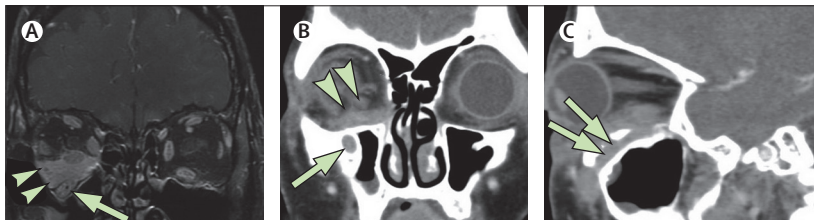


Figure 2: IgG4-related disease of the right inferior orbital nerve in its canal and of the inferior orbital soft tissue (A) Coronal T1 post-contrast MRI, (B) coronal CT, and (C) sagittal CT from the same patient. Arrows point to the right inferior orbital nerve in its canal, and arrow heads point to the inferior orbital soft tissue.

The symptoms of IgG4-related orbital disease depend on the location and extent of disease. In a review⁴² of 172 patients with IgG4-related orbital disease, bilateral disease was observed in 68%, eyelid swelling in 78–100%, proptosis in 29–56%, orbital pain in 17–33%, diplopia in 12–33%, and reduced visual acuity in 20–40%. The lacrimal gland, not technically part of the orbit, is the most common structure of the ocular adnexa to be involved. Lacrimal gland disease is typically bilateral.^{17,42} Extra-ocular muscle enlargement seldom occurs in isolation, being accompanied most commonly by lacrimal gland disease,¹⁷ and is usually not associated with either diplopia or restrictions in ocular movements but rather with ocular prominence (eg, through eyelid swelling) or proptosis that is generally painless. The disease might have a particular predilection for the lateral rectus muscle, but any of the extra-ocular muscles can be involved.¹⁶ Branches of the trigeminal nerve traversing the orbit are often involved.

Enlargement of the infra-orbital nerve (figure 2) strongly suggests the possibility of IgG4-related orbital disease but is almost always an asymptomatic radiographic finding.^{15,43} Pathological sections of these involved nerves have revealed infiltration of IgG4-RD into the epineurium.⁴⁴ Orbital apex lesions afflicting the optic nerve and reducing vision have been described, often appearing as enlarged optic nerve sheaths.^{14,44} Scleritis and nasolacrimal duct disease with secondary epiphora can also occur in IgG4-related orbital or peri-orbital disease, thereby mimicking granulomatosis with polyangiitis (formerly Wegener's granulomatosis).¹⁷

Radiographically, IgG4-RD enhances homogeneously on post-contrast imaging and is typically isointense or hypointense on T2 MRI sequencing, which serves to distinguish IgG4-RD from other diseases that infiltrate the orbit.¹⁵ This hypointensity probably reflects the hypercellularity and fibrosis that is characteristic of IgG4-RD pathology. Moreover, a key distinguishing feature of IgG4-related orbital disease is the involvement of the infra-orbital nerve.^{15,43} Destruction of adjacent bone is not a classic feature of IgG4-related orbital disease^{16,45} but has been described in some cases of IgG4-related sinus disease.^{17,46}

Clinical and imaging features of meningeal disease

IgG4-RD is among the most common causes of non-malignant meningeal inflammation.⁴⁷ IgG4-RD should be considered in cases with unexplained radiological evidence of pachymeningeal disease, particularly if hypertrophic or nodular in appearance. Meningeal IgG4-RD commonly occurs in isolation without evidence of IgG4-RD in other organs and in the setting of normal or only mildly increased serum IgG4 concentrations. Meningeal IgG4-RD manifests mainly in the pachymeninges (ie, dura matter) of the brain or spinal cord.⁴⁸ A case series⁴⁹ of eight patients with systemic IgG4-RD included three patients with

pachymeningitis. By contrast, a single-centre cohort of 125 patients with IgG4-RD included only three (2%) cases of IgG4-related pachymeningitis.⁸ This discrepancy is probably largely due to differences in diagnostic work-up strategies, where in the former study a whole-body PET-CT was performed and thus asymptomatic pachymeningitis was diagnosed, whereas in the later study no screening imagery was conducted. Thus, the true prevalence of pachymeningitis in IgG4-RD is likely high and close to 35% but this is based on a small sample size.⁴⁹

Symptoms of IgG4-related pachymeningitis depend on the location of the disease and occur through several potential mechanisms. First, if the supratentorial dura is involved, traction of the dura leads to pain referred to the distribution of the trigeminal nerve such as the forehead or behind the eye. Second, involvement of the infratentorial dura, which is supplied by branches of the upper cervical roots, causes the referral of pain to the vertex, ear, occiput, or upper neck. Third, meningeal inflammation might encase or compress a nearby structure, typically a cranial nerve. Fourth, blockage of CSF flow at the foramina of Luschka and Magendie can lead to hydrocephalus (figure 3).

A sizeable proportion of patients, even those with extensive hypertrophic pachymeningitis, are relatively asymptomatic, probably explained by a slower disease process. Headache, the most common symptom of IgG4-related meningeal disease, can be of the chronic, intractable variety. A sudden escalation in headache severity should raise suspicion of hydrocephalus, particularly if the headache has a positional quality. IgG4-related pachymeningitis involving the dura around the cavernous sinus can present with symptoms of Tolosa-Hunt syndrome, such as proptosis, diplopia, and retro-orbital pain (figure 4).

Other middle cranial fossa dural lesions can present with visual acuity problems or visual field loss due to compression of the optic nerve and chiasm.⁵⁰ Focal motor weakness, sensory loss, and seizures can also result from cortical compression by pachymeningeal disease, and disease of the basal dura can present with multiple successive cranial nerve palsies.⁵¹ Manifestations of vestibulocochlear nerve palsy occur as progressive asymmetric deafness, vertigo, and tinnitus and can eventually lead to impaired equilibrium and gait. Facial nerve palsy can also occur with basal pachymeningitis and is frequently mistaken for Bell's palsy.^{52,53}

Rare reports exist of IgG4-RD affecting the leptomeninges (ie, pia matter) in the absence of pachymeningeal involvement. IgG4-RD in the leptomeninges is detected with MRI by extension of the disease into the sulcal spaces.^{39,40} These patients typically present with subacute dementia, which is important to identify because it is treatable.^{39,40,54} The dura matter of the spinal cord can also be involved with IgG4-related pachymeningitis,^{55–58} leading through its

mass effect to a myelopathy, radiculopathy, or a myelodradiculopathy syndrome.

Radiographically, IgG4-related pachymeningitis typically induces a homogeneous hypertrophic and nodular pattern of meningeal thickening that is difficult to differentiate from other causes of hypertrophic pachymeningitis such as granulomatosis with polyangiitis, Erdheim-Chester disease, sarcoidosis, lymphoma, and tuberculosis (panel).^{47,54} Results of non-contrast CT studies show IgG4-related pachymeningitis as isodense lesions. Similarly, data from non-contrast MRI studies reveal these lesions to be hypointense or isointense relative to the brain parenchyma on T1-weighted and T2-weighted imaging (figure 5).^{45,59} Both CT and MRI imaging modalities show homogeneous enhancement on post-contrast studies, with lesions that enhance gradually on dynamic studies.^{45,59}

Intralesional calcifications and adjacent bone destruction are atypical.⁴⁵ IgG4-related pachymeningitis is typically not demonstrable with ¹⁸F-fluorodeoxyglucose PET (FDG-PET)⁴⁹ because of the high degree of uptake by normal underlying brain tissue but is more readily detected with ¹¹C methionine PET (MET-PET), which has low uptake in the normal brain.⁶⁰

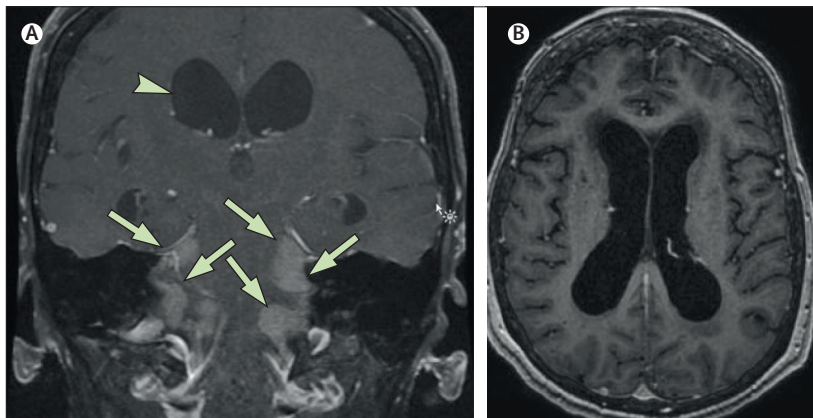


Figure 3: Hydrocephalus resulting from CSF obstruction at the foramina of Luschka and Magendie (A) Coronal T1 post-contrast sequence showing IgG4-related pachymeningitis (arrows) causing obstruction of CSF outflow at the level of the foramina of Luschka and Magendie with secondary hydrocephalus (arrow head). (B) Axial T1 post-contrast slice from the same patient showing more evidence of hydrocephalus.

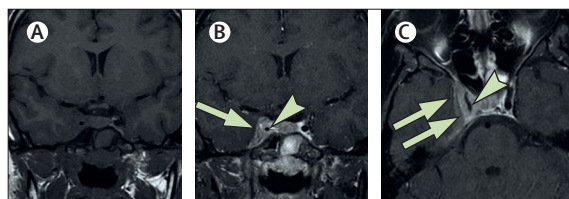


Figure 4: Right cavernous sinus IgG4-related disease mimicking Tolosa-Hunt syndrome with secondary compression of the right internal carotid artery (A) Coronal T1 pre-contrast MRI sequences, (B) post-contrast MRI sequences, and (C) axial T1 post-contrast MRI sequences from the same patient. Arrows indicate the affected cavernous sinus and arrow heads point out the right internal carotid artery.

Panel: Differential diagnosis of hypertrophic pachymeningitis

Connective tissue disease

- IgG4-related pachymeningitis
- Granulomatosis with polyangiitis
- Rheumatoid arthritis
- Sarcoidosis
- Giant cell arteritis
- Behcet disease
- Sjogren syndrome

Infectious

- Mycobacterium tuberculosis
- Syphilis
- Lyme Disease
- Cysticercosis
- Fungal

Neoplastic

- Lymphoma
- Dural carcinomatosis
- En-plaque meningioma
- Erdheim-Chester disease

Other

- Mucopolysaccharidoses
- Intracranial Hypotension

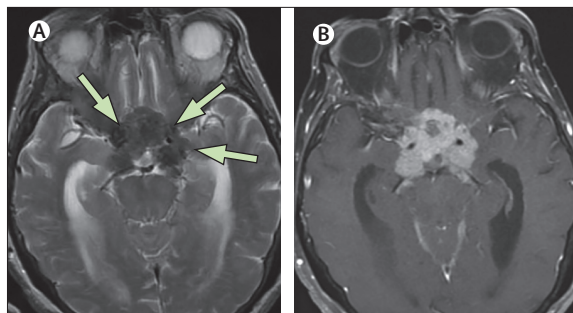


Figure 5: Key radiographic features of IgG4-related disease highlighted in a patient with IgG4-related pachymeningitis of the basal dura. (A) Axial T2 sequence highlighting the hypo-intense radiographic feature of IgG4-RD on MRI (arrows). (B) The corresponding cut on axial T1 post-contrast in the same patient highlighting the homogeneous enhancement of IgG4-RD on contrast-enhanced imagery. Arrows point to the pachymeningeal disease.

Clinical and imaging features of pituitary gland and stalk disease

IgG4-RD commonly involves the pituitary gland and stalk in what is collectively known as IgG4-related hypophysitis.⁴⁸ This inflammatory process can be classified further into anterior gland disease, posterior gland with stalk disease, and panhypophysitis, according to the anatomical structure involved and the resulting clinical picture. Panhypophysitis is the most common presentation.^{10,23} Similar to IgG4-related pachymeningitis, IgG4-related hypophysitis occurs commonly as a single-organ manifestation of the disease rather than in the context of

multiorgan IgG4-RD. IgG4-related hypophysitis was the aetiology of hypopituitarism or central diabetes insipidus in seven (4%) of 170 consecutive patients with those conditions, but accounted for 30% of those cases with MRI evidence of hypophysitis, defined as a homogeneously enhancing enlarged pituitary gland or stalk.¹⁰

Patients with IgG4-RD that primarily affects the anterior pituitary gland tend to present with reduced libido, hypogonadism, hypothyroidism, and hypoadrenalism. Hormonal disturbances of hypopituitarism are more common in IgG4-related hypophysitis than in lymphocytic hypophysitis, an important differential diagnosis.^{11,12} Patients with IgG4-RD of the posterior pituitary gland or infundibular stalk present with symptoms of central diabetes: polyuria and polydipsia.⁶¹ Hypopituitarism has been reported to occur in up to 83% of IgG4-related hypophysitis cases,^{10,11} diabetes insipidus in up to 72%,^{10,11} and both syndromes in 59%.¹⁰

Discrepancies between the radiological findings and the true histopathological extent of the disease sometimes lead to clinical features not predicted by imaging. Constitutional symptoms such as headache, fever, weight loss, appetite loss, lethargy, hypersomnia, and fatigue also occur. Clinical outcomes vary; many patients do not regain pituitary function and need long-term hormonal replacement therapy despite radiographic resolution of the pituitary lesions after treatment; others, however, return to a functionally normal endocrine status after successful treatment.

MRI shows an enlarged pituitary gland or thickened stalk with hypointense T2 characteristics that enhance homogeneously with gadolinium (figure 6).^{45,59} Cystic formations inside the enlarged gland can be detected.¹⁰ Empty sella syndrome after glucocorticoid treatment has been described.¹⁰ Loss of the normal posterior pituitary gland and stalk T1 precontrast hyperintensity has also been observed.^{10,45,59} The imaging features of IgG4-RD involvement of the pituitary gland and stalk disease mimic those of sarcoidosis, granulomatosis with polyangiitis (formerly Wegener's granulomatosis), Langerhans cell histiocytosis, syphilis, tuberculosis, and lymphocytic hypophysitis. PET imaging of the pituitary region is generally of low utility secondary to the masking of tracer-uptake by the brain parenchyma.^{49,62} On the other hand, whole-body PET imaging is often useful in the identification of extra-CNS IgG4-RD or to identify disease patterns in other organs that are more suggestive of other diagnoses.

Clinical and imaging features of peripheral nerve disease

Peripheral nerves can be involved in IgG4-RD by one of two mechanisms. First, the nerves can be compressed by the mass effect of local involvement of non-neural tissue.⁶³ Second, the epineurium might be involved when IgG4-RD causes chronic inflammatory aggregates, with or without further infiltration into the perineurium.^{44,64} Both mechanisms can result in varying degrees of axonal damage, with ensuing loss of both large and small

myelinated fibres, with or without demyelination.⁶⁴ Inflammatory IgG4-RD cell infiltration into the endoneurium or nerve fibres on histopathology has never been reported. Thus, the axonal loss is probably related to functional trophic changes stemming from chronic inflammation.

IgG4-related neuropathy might be underdiagnosed, as suggested in a review of 149 consecutive patients previously diagnosed with a different inflammatory neuropathy on nerve biopsy. When biopsies were stained for IgG4-RD, 29 patients, 11 of whom had been diagnosed with Churg-Strauss and 11 with microscopic polyangiitis vasculitic neuropathies, had IgG4-plasma cell infiltrate in sufficient quantity to fulfil diagnostic criteria for IgG4-RD.⁶⁵ In another study, extensive IgG4-related neuropathy was reported for the visceral nerves of the kidney, prostate, epicardium, abdominal aorta, retroperitoneum, mesentery, and, most extensively, the nerves comprising Auerbach's plexus of the intestinal wall.⁶⁶

Indeed, retroperitoneal resections to treat presumed malignancies often reveal encasement of peripheral nerves within the inflammatory infiltrate of IgG4-RD. This might be one mechanism for the intense pain experienced by many patients with IgG4-related retroperitoneal fibrosis. Asymptomatic spinal nerve root involvement has been reported,⁴⁴ but no accounts exist for IgG4-RD of the dorsal root ganglia, the brachial and lumbosacral plexuses, or isolated IgG4-related small-fibre neuropathy. Imaging typically shows focal round or lobular lesions that enhance homogeneously, surrounding the affected nerve, but not leading to calcification or necrosis.⁴⁴ These lesions are FDG-avid on PET scans.⁴⁴

Parenchymal brain disease

Brain tissue is rarely involved by IgG4-RD, but isolated case reports exist. We are aware of only one case with sufficient evidence for true parenchymal brain disease.⁶⁷ Two reports exist of IgG4-related pachymeningitis in which the underlying brain parenchyma was suggested to also be involved with IgG4-RD based on radiographic findings.^{68,69} In these two cases, the brain was not biopsied; additionally, the proximity of the parenchymal lesions to, and their concomitant resolution with, the overlying IgG4-related pachymeningitis after glucocorticoid therapy could indicate that the parenchymal radiographic lesions are inflammatory findings due to mechanical compression by the pachymeningeal disease, rather than true IgG4-RD of the brain parenchyma. Finally, an intraventricular IgG4-RD mass lesion, proven by biopsy to be IgG4-RD, has been described in one report.⁷⁰ Spinal cord parenchymal disease has never been reported, but tissue sampling from this region is not conventional.

Diagnosis

The diagnosis of IgG4-RD of the nervous system, similar to the disease overall, requires consideration of clinical, radiographic, serological, and pathological evidence.



Figure 6: IgG4-related pituitary and stalk disease

Sagittal T1 post-contrast MRI showing that the enlarged pituitary gland extends well beyond the boundaries of the sella turcica. The arrows point to pituitary disease, and the arrow head points to stalk disease.

Serology

A substantial increase in serum IgG4 concentration is useful in suggesting the diagnosis of IgG4-RD but is not sufficient for diagnosis. About 30% of patients with histopathologically confirmed disease have normal serum IgG4 concentrations. Additionally, among patients with increased serum IgG4 concentrations, the range of values is extremely broad, ranging from more than 50 times the upper limit of normal (135 mg/dL) in some patients with multiorgan disease to normal in many patients with limited or single-organ disease. The factors contributing to such variability in the IgG4 response are poorly defined. IgG4 concentrations correlate only loosely with disease activity and are imperfect predictors of the need for additional treatment.

In a meta-analysis⁷¹ of more than 1200 patients with IgG4-RD and nearly 5700 healthy controls, IgG4 concentrations more than 135 mg/dL as a marker for IgG4-RD had a sensitivity of 87% and specificity of 83%. Steroid therapy significantly lowers IgG4 concentrations in patients with IgG4-RD²⁹ and should ideally be measured before initiating steroid therapy because the concentration will decrease afterwards. Normal total IgG concentrations do not exclude the possibility of increased IgG4 concentration, and a differential subclass quantification is necessary.

Markers of systemic inflammation are generally absent in isolated CNS IgG4-RD, and their occurrence should raise the clinician's suspicion towards the presence of the disease in other organ systems. Increased peripheral plasmablasts¹⁸ have been linked to systemic IgG4-RD but not to CNS disease. Increased serum IgE concentration and eosinophils have also been linked to systemic IgG4-RD and occur more frequently in elderly patients with active multiple-organ involvement.⁸ No antigen-specific autoantibodies have been confirmed to be associated with IgG4-RD.⁹

Cerebrospinal fluid

IgG4-related pachymeningitis typically shows a mild CSF pleocytosis, occasionally including more than 200 nucleated cells per mm.⁹ The pleocytosis is predominately lymphocytic and associated with mild increases in CSF protein concentration. Oligoclonal bands have been reported in cases of IgG4-related pachymeningitis, suggesting intrathecal production of antibodies.⁷² Completely normal CSF examinations have also been observed.⁹

In one study,⁷² serum IgG4 concentrations were compared for three patients with IgG4-related pachymeningitis, 21 patients with other causes of hypertrophic pachymeningitis, and nine controls. Whereas a CSF IgG4 cutoff of 2.27 mg/dL correctly categorised all three patients with IgG4-related pachymeningitis, only one patient with pachymeningitis of another aetiology and none of the controls had a CSF IgG4 concentration so high. A cutoff of 0.47 for IgG4-Loc (an index for intrathecal IgG4 production) diagnosed all the patients with IgG4-related pachymeningitis and none of the patients in the other two groups. The three patients with IgG4-related pachymeningitis in this study had normal serum total IgG and IgG4 concentrations, consistent with the idea that patients with isolated intracranial IgG4-RD have an inflammatory reaction compartmentalised within the intrathecal space. Pituitary gland and stalk inflammation are generally not associated with major changes in the composition of the CSF.

Pathology

Histopathology findings and immunostaining studies are important in establishing the diagnosis of IgG4-RD. Immunohistochemical staining for IgG-positive and IgG4-positive plasma cells is not routine in most laboratories, and the clinician will need to alert the pathologist of his suspicion for IgG4-RD.

A dense lymphoplasmacytic infiltrate enmeshed by an irregularly whorled, storiform (from storea, Latin for woven mat) is typical of IgG4-RD. Obliterative phlebitis and tissue eosinophilia are also present in many cases.⁷³

Search strategy and selection criteria

We searched PubMed for relevant case reports, case series, original research, and reviews in English that were published between Oct 1, 2001, and Oct 31, 2017. Different combinations of the following search terms and their synonyms were used: IgG4-related disease, immunoglobulin-G4, nervous, cerebral, encephalopathy, intracranial, stroke, cranial nerve, pachymeningeal, hypophysitis, hypopituitarism, orbital pseudotumour, skull, myelopathy, cauda equina, neuronopathy, neuropathy, perineuritis, myopathy, and myositis. References within the identified papers were also screened for further potentially relevant reports.

IgG4-related hypophysitis commonly shows storiform fibrosis but rarely obliterative phlebitis.^{10,62} Anterior pituitary gland biopsy will often show residual glandular epithelial cells.^{11,74}

A substantial proportion of plasma cells (typically >40%) stain positively for IgG4.⁷³ However, neither the number of IgG4-positive plasma cells per high-power field nor the ratio of plasma cells positive for IgG4 and IgG is specific for the diagnosis of IgG4-RD. Immunostaining findings must be interpreted in view of the histopathology features and in the context of clinical and radiological findings. Reliable diagnoses can seldom be made on the basis of pathology findings alone.

Treatment approaches

An international consensus statement from more than 40 investigators from Europe, North America, and Asia (considered the best available evidence) recommends glucocorticoids as first-line treatment for systemic IgG4-RD, initially as a monotherapy.⁷⁵ With neurological complications, however, the risk of severe and irreversible disease-related damage from incompletely treated IgG4-RD is high. Moreover, the morbidity from lengthy courses of high-dose glucocorticoids is substantial, and glucocorticoids often fail to induce durable remissions. Little evidence exists to suggest that conventional glucocorticoid-sparing therapies such as azathioprine and mycophenolate mofetil offer additional benefit beyond that achieved with glucocorticoids, and no consensus exists about the utility of these drugs as adjunct therapies to glucocorticoids.

These considerations often justify the use of B-cell depletion with rituximab (two doses of 1 g intravenously, separated by 2 weeks, repeated at 6 month intervals) in combination with glucocorticoids.⁷⁶ We usually begin treatment for IgG4-RD with the combination of prednisone (40–60 mg per day for 1 month and tapering to discontinuation over 2–3 months) and rituximab. In the presence of serious neurological disease, multiorgan IgG4-RD beyond the nervous system, or increases of serum IgG4 concentrations many times higher than the upper limit of normal for the reference laboratory, strong consideration should be given to the early use of B-cell depletion since glucocorticoid treatment alone is unlikely to lead to satisfactory treatment responses at acceptable risk–benefit ratios from therapy. CNS imaging should be repeated about 3 months after the start of such a regimen to monitor the response. The optimal remission induction regimen for different scenarios of nervous system involvement and appropriate approaches to remission maintenance regimens remain to be defined.

Considerations for future research

The highly oligoclonally restricted repertoires of both plasmablasts and CD4+ CTLs in IgG4-RD strongly suggest an antigen-driven disease. Efforts to identify the antigen (or, perhaps more likely, antigens) that trigger

this disease are an important focus of ongoing research. Continued attempts to elucidate the relationships between cells of the B-cell lineage and CD4+ T lymphocytes (both the CD4+ CTL and the Tfh cell) could provide clues to the functioning of human immunology beyond IgG4-RD and identify new targets for therapy. Clinical trials focusing on the elimination of chronic courses of glucocorticoid therapy might offer important steps forward in treatment and improvement of patients' outcomes and quality of life.

Contributors

The content of this Review was conceived and outlined by MAA, NV, and JHS. MAA and JHS wrote the first draft, and all authors participated in revisions. All authors have read and approved the final version for publication.

Declaration of interests

JHS has received research grants and consulted for Roche/Genentech and Xencor on the topic of IgG4-related disease. MAA and NV declare no competing interests.

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